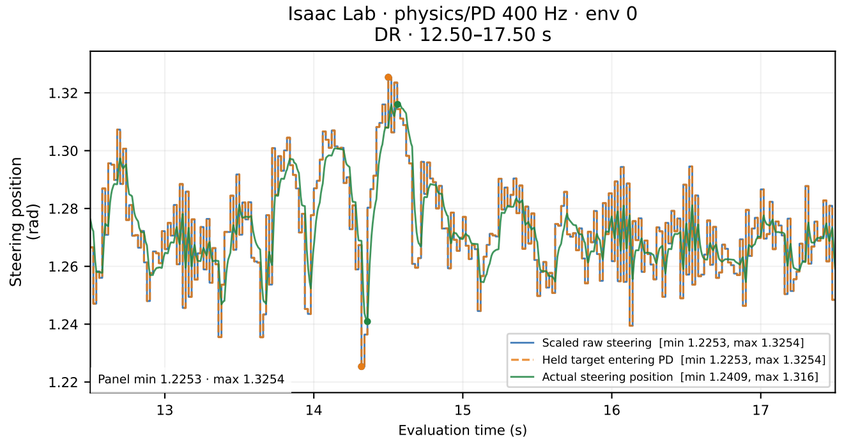
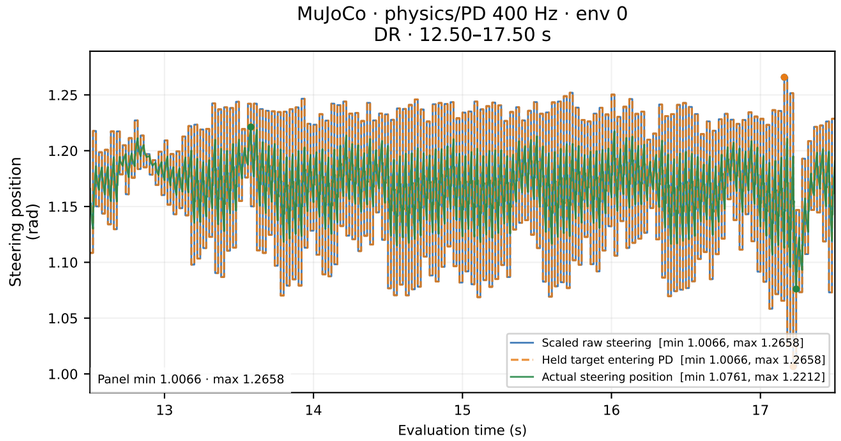
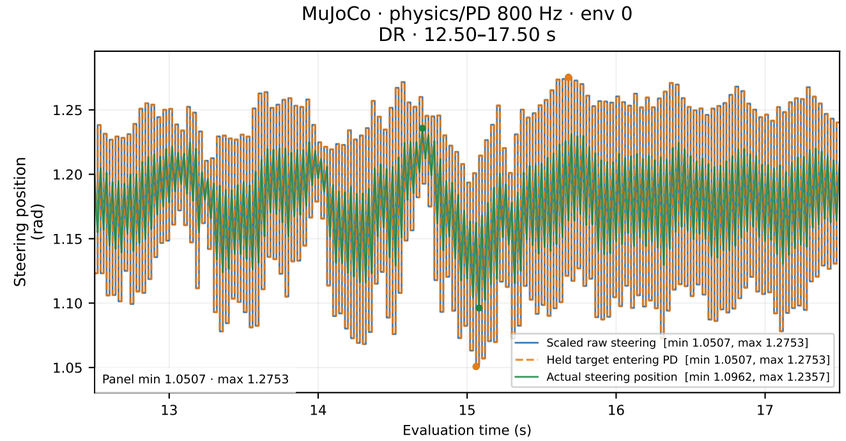
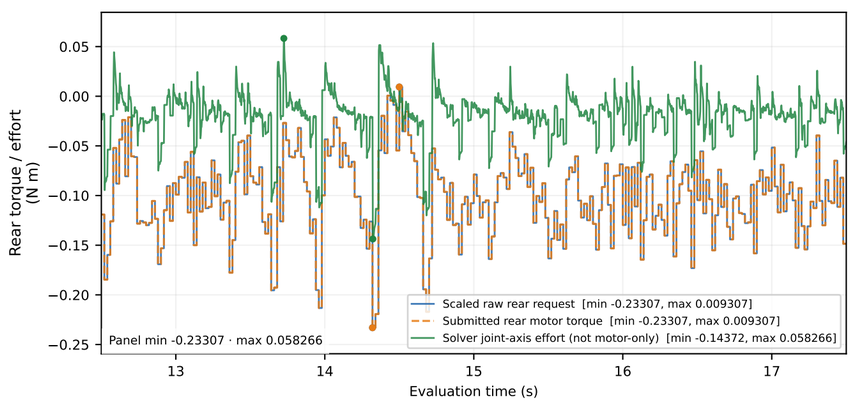
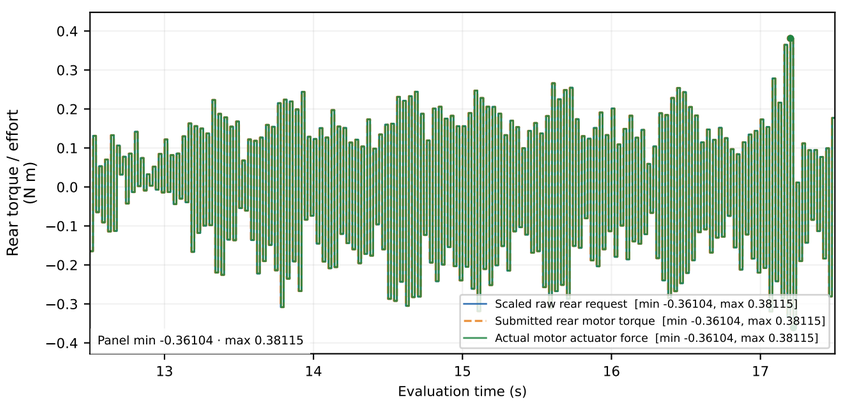
model_2500 · Demand versus actuation · env 0 · five-second detail

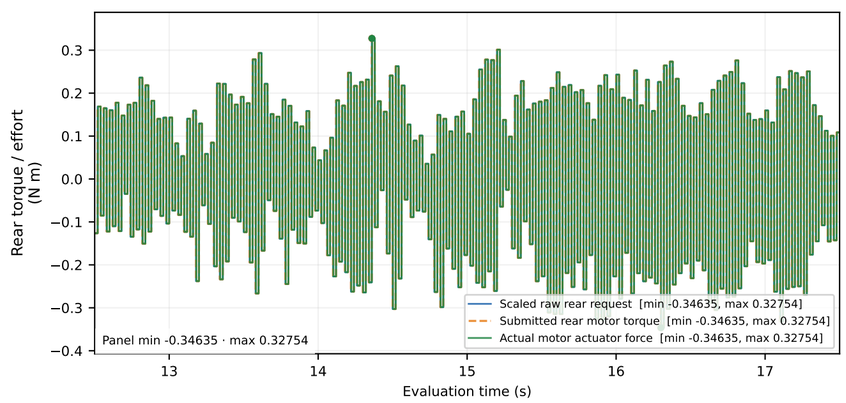












50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Each column shows the central ≤5 s of that environment's longest completed episode; different absolute windows.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 0; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.04518, 0.804024, 0.959366, 0.816123]
  CoM XYZ mm [4.87718, -3.711, 0.621036]
  mu_s [0.741039, 0.681711, 0.592046, 0.532645]
  mu_d [0.614405, 0.611756, 0.534947, 0.532645]
  Kp [steer,rear] [10.6517, 0]; Kd [0.209184, 0]
  spawn roll 0.0198978 rad; rate 0.00125125 rad/s
  gyro bias XYZ rad/s [-0.000195088, -0.000150685, -2.64955e-05]
  gravity bias YZ [0.00115356, 0.00192213]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 0; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05389, 1.02143, 1.02368, 0.92158]
  CoM XYZ mm [-4.69182, -0.632826, -2.85415]
  mu_s [0.666375, 0.640016, 0.669571, 0.71155]
  mu_d [0.519145, 0.513141, 0.669571, 0.71155]
  Kp [steer,rear] [9.46788, 0]; Kd [0.191255, 0]
  spawn roll 0.0628041 rad; rate -0.166514 rad/s
  gyro bias XYZ rad/s [0.000138097, -8.54505e-05, 3.17079e-05]
  gravity bias YZ [0.00125118, -0.000618693]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 0; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05389, 1.02143, 1.02368, 0.92158]
  CoM XYZ mm [-4.69182, -0.632826, -2.85415]
  mu_s [0.666375, 0.640016, 0.669571, 0.71155]
  mu_d [0.519145, 0.513141, 0.669571, 0.71155]
  Kp [steer,rear] [9.46788, 0]; Kd [0.191255, 0]
  spawn roll 0.0628041 rad; rate -0.166514 rad/s
  gyro bias XYZ rad/s [0.000138097, -8.54505e-05, 3.17079e-05]
  gravity bias YZ [0.00125118, -0.000618693]
Six significant digits shown; full precision: sampled_dr_values.json.
```